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Failure of thermoacoustic instability control due to periodic hot gas ingestion in Helmholtz dampers

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ABSTRACT

Passive damping devices such as Helmholtz dampers are often employed in gas turbines to mitigate thermoacoustic instabilities. The periodic ingestion of the combustion chamber hot gas in the resonant cavity of Helmholtz dampers can be a serious issue, if not considered in the design phase of these devices. The periodic hot gas ingestion modulates the density in the damper neck and thereby the damper's resonance frequency. This effect, combined with the nonlinearities in the acoustic damping caused by the periodical reversal of the flow direction in the neck, can prevent the intended control of the thermoacoustic feedback. Despite its relevance for gas turbine design, this topic has not received significant attention in literature. This study presents experimental and theoretical investigations of that problem. A physics-based model is derived with a few empirical parameters, calibrated with data which were collected from an experimental setup comprising a tunable Helmholtz damper connected to a combustor operated at atmospheric pressure. This model of noise-driven coupled oscillators with nonlinear damping and stiffness is capable of reproducing the bistable dynamics observed in the experiments for specific ranges of equivalence ratios. The model is then used to draw general conclusions about the robustness of dampers with respect to this unwanted phenomenon. In particular, the influence of the geometry and flow conditions on the damping and stiffness nonlinearities, which both participate to possible failure of the passive control, is described. We show that in the case of broadband dampers, the effect of the nonlinear stiffness is weaker than the one of damping nonlinearity, and that it can become significant for more narrow-band dampers. The results can be used for designing acoustic dampers that are robust with regard to the risk of spontaneous loss of thermoacoustic stability due to periodic hot gas ingestion.

1. Introduction

Guided from the increasingly stringent pollutant emissions regulations of the last years, the future of the energy sector is oriented towards the integration of gas turbines burning sustainable fuels in combination with renewable energy technologies. In fact, the volatility of renewable energy production could be compensated by employing the surplus power to produce sustainable fuels that could be then burnt on-demand in low-emissions gas turbines [1,2]. For this reason, the attention of gas turbines manufacturers has turned to combustors capable of operating with these fuels while reducing pollutant emissions. The goal can be achieved by burning fuels in lean premixed combustion. This regime reduces the NO_x emissions by lowering the peak temperature in the combustion chamber. The major drawback of these systems is their higher tendency to develop thermoacoustic instabilities [3]. They result from the coherent interaction between the unsteady heat release and the acoustic pressure field [4] and can damage structural

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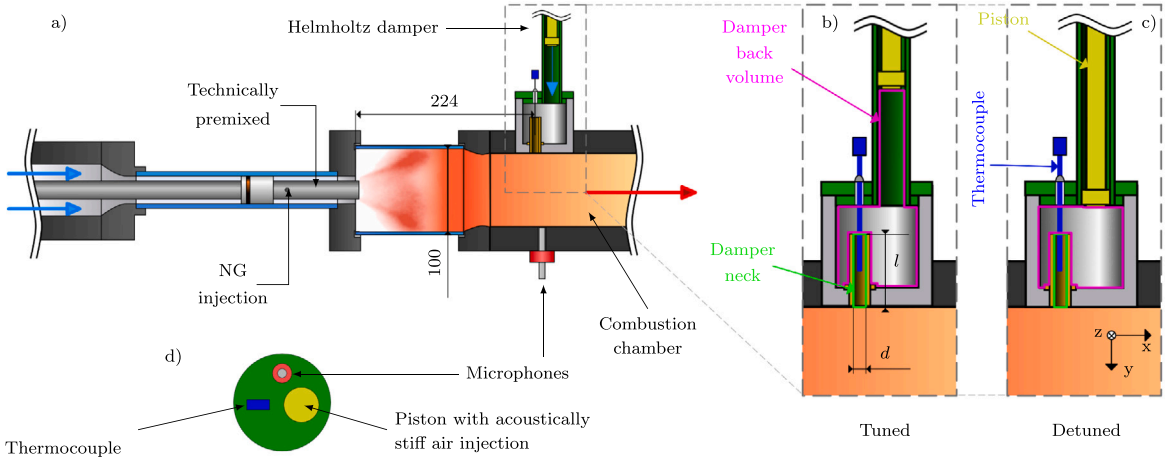

Fig. 1. (a) sketch of the combustor where the main components are indicated with arrows. Subfigures (b) and (c) show a closer view of the Helmholtz damper with the piston in tuned and detuned position, respectively. (d) shows the top view of the Helmholtz damper. The dimensions are in mm.

components, reducing the lifetime of the machine significantly. The prediction and suppression of these self-induced oscillations is a major difficulty in the design of modern gas turbines and aeroengines combustors and it is often achieved by coupling acoustic dampers such as Helmholtz resonators to the combustion chamber, e.g. [5–9].

A Helmholtz damper is schematically illustrated in Fig. 1. It consists of a back volume, connected to a combustion chamber through a neck. A common approach to describe the linear behaviour of these oscillators is through an analogy with a mechanical mass–spring–damper oscillator, where the mass of the system represents the incompressible motion of the mass of air in the neck of the damper and the spring the compressibility of the air in the back volume [10].

The Helmholtz resonance frequency, f_d, is then given by:

$$f_\mathrm{d} = \frac{\omega_\mathrm{d}}{2\pi} = \frac{c_\mathrm{d}}{2\pi}\sqrt{\frac{a}{V_\mathrm{d} l_\mathrm{eff}}}, \tag{1}$$

where c_d denotes the speed of sound in the damper, a the cross-sectional area of the neck, V_d the back volume of the damper, and l_eff the effective length of the neck which is the sum of the physical length of the neck and the end-correction at both ends of the neck [11].

The description of the behaviour of a damper in isolation, is however not sufficient to analyse its effect on the system's stability when it is coupled to a chamber. Among the numerous articles dealing with the stability of acoustic modes in systems composed of dampers coupled to a cavity, there is notably the work of Fahy and Schofield [12]. They investigated the damping enhancement of one eigenmode of a chamber induced by the presence of a well-tuned damper. This linear theory was then extended by Cummings [13] for multiple cavity modes.

Over the last decades, one of the key themes in the community of researchers dealing with thermoacoustics of combustors has been the investigation of passive control approaches of the instabilities using dampers [14–17]. In particular, several studies have described the effect of placement and different geometrical parameters on the damping effectiveness of Helmholtz resonators. Mensah and Moeck [18] proposed a computationally-efficient adjoint-based method to increase the effectiveness of acoustic dampers by finding the optimal positioning and tuning on a generic annular combustor model. The influence of the location of Helmholtz dampers on the modal dynamics of annular combustion chambers was also investigated by Mazur et al. [19]. Betz et al. [20] studied experimentally how the number of quarter wave resonators, their spatial distribution and the amount of purge air affect the damping characteristics of an annular lab-scale burner. Yang et al. [21] presented a linear low-order model describing the acoustic behaviour of a hot annular duct with a mean flow, accounting for the presence of Helmholtz dampers. Furthermore, they introduced a gradient-based optimization method of the longitudinal and axial positions and dimensions of dampers in order to obtain the highest damping of acoustic modes in a given frequency range. In addition, Zalluhoglu et al. [22] focused on dampers placement on a Rijke tube, Jo et al. [23] on the orifices dimensions and number of cavities and Pan et al. [24] on the presence of grazing flow.

In gas turbine applications, Helmholtz dampers are usually purged with bypass flow coming from the compressor (purge flow) in order to avoid the penetration of the hot and corrosive combustion products in the back volume of the damper and to ensure a large range of acoustic amplitude for which the damping of the device remains linear. When the amplitude of the acoustic velocity is larger than the mean velocity of purge flow in the neck of the damper, hot gas is periodically ingested by the damper neck. The periodic reverse flow of hot gas in the neck has two effects. First, it dynamically varies the density in the neck of the damper, affecting the resonance frequency. As a consequence, it perturbs the coupling with the combustion chamber by introducing a nonlinearity in the reactance of the damper. Second, it influences the acoustic dissipation occurring at the neck due to viscous effects in a nonlinear

way, i.e. the Helmholtz damper resistance becomes nonlinear with respect to the acoustic amplitude when the latter is large enough to induce periodic flow reversal in the neck.

A naive solution to mitigate the risk of damping failure due to dynamic hot gas ingestion would be to simply increase the purge flow velocity in the neck of the damper. However, this has to be done carefully, as it is not beneficial from other key perspectives. First of all, there is an optimum mass flow [25] which should not be exceeded, because it would otherwise worsen the linear stability of the thermoacoustic system. Second, any increase of the purge mass flow of the damper has a negative effect on the global cycle efficiency of the gas turbine, as the compressor air used for this purpose does not contribute to the power generation of the cycle. For these reasons, the importance of not oversizing the purge mass flow, combined with the potential severe damages induced by the phenomenon, make the correct modelling and prediction of the hot gas ingestion in Helmholtz dampers an important research topic. Furthermore, only a few research papers deal with this phenomenon and there is a gap in that regard in the literature. This work aims at disentangling the two different types of nonlinearities in order to better design Helmholtz dampers for suppressing thermoacoustic instabilities.

In the next paragraphs, we briefly review important contributions which are relevant to the design of robust and broadband Helmholtz dampers for gas turbines applications and which partly address the mechanism involved in the nonlinear dynamics of periodic hot gas ingestion events.

The impedance model of a Helmholtz resonator was originally introduced by Helmholtz [26] and, among many others, was further developed by Keller and Zauner [27] and Bellucci et al. [10]. Indeed, the latter reference presents a model of the impedance of a Helmholtz resonator which accounts for the purge flow and the nonlinearity induced by the periodic reversal of the flow for the case of damper necks that are subject to mean bias flow in isothermal configurations. Subsequent studies further developed this model by considering physical constraints pertaining to combustion environment. In particular, Ćosić et al. [28] extended the nonlinear damper model of Keller and Zauner [27] to account for the effect of the periodic ingestion of hot gas in the neck of a Helmholtz resonator. The density variation was modelled with a time-dependent and spatially-averaged equation. The amount of ingested hot gas was related to the acoustic oscillations in the combustion chamber at the damper axial location. When compared against experimental results, this nonlinear model allows predictions of the shift of the resonance frequency associated with the change of the mean air temperature in the neck induced by periodic hot gas ingestion. Their work focused on the nonlinear response of the damper to acoustic forcing. They however did not investigate the coupled dynamics of the system composed by the damper and the combustion chamber. Later, Bothien et al. [29] analysed the drift of resonance frequency caused by a neck end-correction which depends on the temperature difference between the damper and the combustion chamber for acoustic forcing amplitudes that are sufficiently low to not induce periodic flow reversal, i.e. corresponding to a linear regime. They introduced a correction factor dependent on the density ratio between the back volume of the damper and the combustion chamber. Wassmer et al. [30] subsequently tested this length correction model against experimental data obtained with a Helmholtz damper connected to a combustion chamber. They showed that it overall improves the resonance frequency estimates compared to the isothermal model when temperature differences are not too large. Yang and Morgans [31] also investigated and modelled the effect of the difference of temperature between air in the neck of a Helmholtz resonator and the combustion chamber for acoustic forcing amplitudes that are sufficiently low to be in the linear regime. Ćosić et al. [32] built upon [28] and further elucidated the effect of the interaction between the purge flow and the hot grazing flow in a Helmholtz damper on the damping efficiency and the resonance frequency of the device. In particular, they scrutinized the effect of the acoustic forcing amplitude on the resonance frequency. Zahn et al. [33] introduced a numerical method to predict the linear stability limit of an annular combustor when coupled or not to quarter waves tubes. The validation of their model against experimental results required a sensitivity analysis to account for the damping performance deterioration induced by the damper resonance frequency shift caused by hot gas ingestion. More recently, Betz et al. [34] experimentally analysed the relation between the mean temperature in the necks of a Helmholtz resonator with multiple necks and the amplitude of the acoustic pressure in an annular combustion chamber. They reported ingestion of hot gas in some necks of the resonator, leading to a detuning and, consequently to a reduction of their damping performance.

The main novelty of the present work is that we model the nonlinear two-way interaction between a Helmholtz damper and the thermoacoustic dynamics of a combustor, while previous studies were limited to the one-way nonlinear response of the damper to imposed acoustic forcing. Indeed, we investigate experimentally and theoretically the nonlinear dynamics of Helmholtz dampers undergoing spontaneous periodic hot gas ingestion due to their interaction with a thermoacoustic mode of the combustion chamber.

The coupling between the damper and the combustion chamber is modelled following the work of Bourquard and Noiray [25]. As previously mentioned, they showed analytically and validated experimentally how to optimize the purge massflow of a Helmholtz damper to get the best stabilization of a linearly unstable thermoacoustic mode, i.e. tuning the damper such that the coupled system damper/combustor is brought to one of its exceptional points. An exceptional point (EP) in the present case of a pair of coupled oscillators is a spectral degeneracy obtained by tuning the system parameters: At the EP, not only the pair of eigenvalues, but also their associated eigenmodes, coalesce. When moving away from the EP by changing a parameter of the system, at least one of the eigenvalues has a larger growth rate, which means that the system is less linearly stable.

They validated their model against experiments, in which thermoacoustic instabilities were mimicked with electro-acoustic instabilities. In a subsequent work [35], they considered the same experimental system, i.e. a purged damper coupled to a self-sustained electro-acoustic eigenmode of a chamber and extended their model to account for periodic flow reversals in an isothermal configuration. In their work a modal projection was considered and the self-sustained oscillation of the dominant eigenmode was modelled with a Van der Pol oscillator [36], which will be done also in the present work. Building upon the knowledge gained from the isothermal electro-acoustic experiments and modelling results of [25,35] we will perform thermoacoustic experiments using a combustor equipped with a damper purged by a cold flow and derive a low order model of the corresponding nonlinear stochastic

dynamics. We will show that during the transition from thermoacoustic resonance driven by turbulence-induced stochastic forcing to thermoacoustic limit cycle, which occurs when the bifurcation parameter is slowly increased, intermittency can be observed in the form of erratic bursts of high acoustic amplitude. Intermittency of thermoacoustic system is the topic of a recent review [37] and the interested reader can for example refer to [38] for a study on thermoacoustic intermittency induced by stochastic fluctuations of the time delay in the response of a turbulent flame to acoustic forcing. Here we unravel another possible root cause of intermittency: the periodic hot gas ingestion in a damper connected to a turbulent combustor.

The paper is organized as follows. First, in Section 2, the experimental set-up and results will be introduced. The acoustic pressure oscillations in the combustion chamber are analysed when the damper is tuned to or detuned from the dominant thermoacoustic mode eigenfrequency. In Section 3, the model is presented and then validated against the experimental results in Section 4. Finally, the model is employed to study variations of the geometry of the neck and isolate the effect of the hot gas ingestion on the coupled system.

2. Experiments

Experiments were conducted to investigate the effect of the hot gas ingestion on a damper-combustion chamber system. This section outlines the experimental test rig, diagnostics and analyses the observations.

2.1. Experimental facility

The experiments are performed on a modular combustion test rig operated at atmospheric conditions (sketched in Fig. 1), similar to the one employed in [39]. The combustion chamber is composed by a round quartz tube having a diameter of 100 mm and a length of 140 mm followed by 2 square cross-section modules of side 88 mm and of length 250 mm. The transition piece between the quartz tube and the modules provides a smooth geometrical transition at constant cross section. The combustor is run in technically premixed configuration of natural gas (NG) and air, i.e. the NG is injected through small holes in the burner just downstream of the swirler. The NG mass flow ($\dot{m}_{\mathrm{NG}}$) is varied between 0.44 and 0.53 g s^{-1}. The adjustable air massflow makes the equivalence ratio (ϕ) range between 0.76 and 0.86. The power and equivalence ratio boundaries allowed to observe the transition from stable to unstable thermoacoustic conditions.

The neck of a Helmholtz damper is located 224 mm downstream of the flame. The back volume of the damper can be varied by adjusting the position of a piston. The resonance frequency of the device ranges between 150 and 350 Hz. The neck of the damper has a length of 40 mm and a cross-sectional diameter of 12 mm. Despite the fact that the damper back volume exhibits a complex shape, one can safely consider it as acoustically compact because its largest dimension of 0.35 m when the piston is completely recessed is significantly smaller than the wavelength of 1.7 m corresponding to the oscillation frequency of the thermoacoustic mode of interest (200 Hz). As shown in Fig. 1, the neck of the damper is a tube which protrudes inside the back volume [40]. The damper was designed to allow for the installation of a thermocouple which can slide in the neck and for enabling a piston-based adjustment of the back volume. The cold purge air enters the damper through small orifices located in the head of the piston. For the purge mass flows considered in this work, the flow at these small orifices is choked, which means that this stiff injection decouples acoustically the air feeding line connected to the damper.

Each combination of NG massflow and ϕ was run at stationary condition, in order to record long acoustic pressure oscillation signals with seven G.R.A.S. 46 BD 1/4″ CCP microphones (sampling rate 5000 Hz). Three of them were connected upstream of the burner (in the combustor plenum), three downstream of the flame (one at the same axial location as the neck of the damper) and one in the back volume of the damper. In order to measure the mean temperature in the neck of the damper, a K-type thermocouple (sampling rate 60 Hz) is inserted and its tip is positioned at its middle (as sketched in Fig. 1).

In the first set of experiments, the damper was tuned, by adjusting the piston position, close to the frequency of the dominant mode of the combustor, around $f_{\mathrm{c}} = 196$ Hz. The purge massflow through the neck of the damper was fixed to 0.5 g s^{-1}, corresponding to a mean velocity in the neck of 4.2 m s^{-1} and which is small compared to the air mass flow through the combustor that participates in the combustion process (between 9 and 12.5 g s^{-1}). This value was chosen as a trade-off that could allow to control the amount of ingested hot gas for the operating conditions considered in this work. In fact, to observe the onset of the hot gas ingestion, the mean flow velocity in the neck has to be larger than the acoustic velocity for some operating conditions for which there is no hot gas ingestion. In the second set of experiments, the damper was detuned with respect to the dominant thermoacoustic mode by changing the piston position. These experiments, for which the resonance frequency of the damper was close to 315 Hz, were used as a benchmark to identify the effect of the damper on the coupled system for configurations where the same amount of damper purge air was flowing inside the combustion chamber.

2.2. Experimental results

The amplitude of the dynamic pressure at different combinations of NG massflows and equivalence ratios are now presented. Plots (a) and (b) of Fig. 2 show the RMS of the acoustic pressure for the tuned and detuned damper conditions. The acoustic signal in the combustion chamber at the axial location of the damper is reconstructed with the most energetic modes of the proper orthogonal decomposition (POD) [41] from the dynamic pressure signals measured with microphones at 6 positions along the combustor. The POD postprocessing allows us to isolate the acoustic component of the dynamic pressure: the application of this method to several microphones signals eliminates the pseudonoise produced by flow structures in the vicinity of each microphone part, which are not

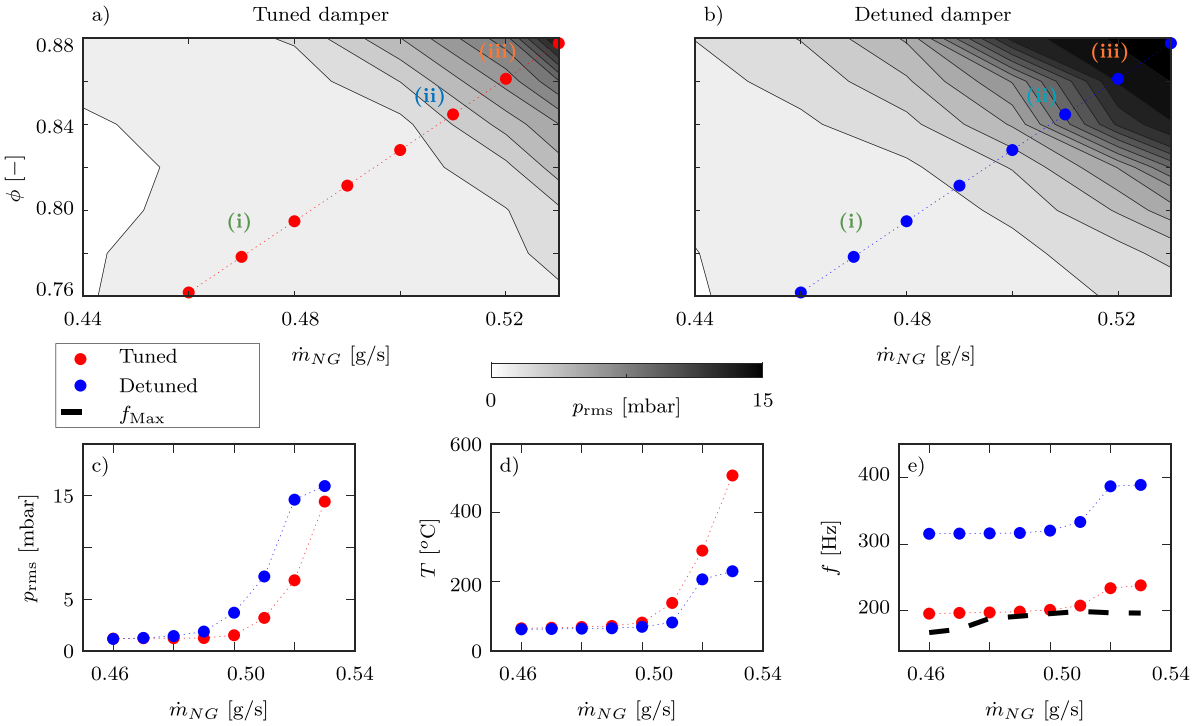

Fig. 2. Contourplots representing the acoustic pressure rms in the combustion chamber at the same axial position as the damper. Different NG massflows ($\dot{m}_{\mathrm{NG}}$) and equivalence ratios (ϕ) were mapped for the configuration with a tuned (a) or detuned (b) damper. The blue and the red dots correspond to the investigated line with constant air considered to analyse the transition from linearly stable to unstable thermoacoustic conditions. Plots (c)–(e) compare respectively the rms of the acoustic pressure p_{rms}, the time averaged temperature in the neck T and the resonance frequency of the damper for the operating conditions at constant air massflow f. The dotted line in (e) indicates the frequency of the dominant peak in the PSD of the microphone signal located in the combustion chamber f_{peak}. Examples of these PSDs are shown in Fig. 3 for the operating conditions (i), (ii) and (iii) indicated in (a) and (b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

correlated to relevant acoustic scales, and which are therefore not contributing to these most energetic POD modes. The acoustic signal is then filtered around its dominant frequency around 196 Hz using a rectangular window function with a bandwidth of 40 Hz. Apart from the damper volume defined by the piston position, the two cases presented in Fig. 2 correspond to identical operating conditions. The mapping of the operating conditions was performed by varying the NG massflow between 0.44 and 0.52 g s^{-1} in increments of 0.01 g s^{-1} and ϕ from 0.76 to 0.86 in increments of 0.01. When the damper's resonance frequency is tuned close to the frequency of the dominant thermoacoustic mode of the combustor, the rms of the acoustic oscillations in the combustor is generally lower than in the detuned case. This observation shows that the tuned damper widens the range of conditions for which the system is thermoacoustically stable. Both tuned and detuned configurations show an increase of acoustic pressure amplitude for high power and equivalence ratios. The broadening of the linearly stable operating window is "only" a few percent of the nominal value of the bifurcation parameter (for example, in Fig. 2c) we see that the rise of the rms level occurs for $\dot{m}_{\mathrm{NG}} = 0.49$ and $\dot{m}_{\mathrm{NG}} = 0.5$ in the detuned and tuned case respectively. It is important to mention that such "limited" effectiveness of the damper in stabilizing the system was intentional: on purpose, we limited the damper volume to be able to observe the hot gas ingestion phenomenon on the considered range of operating parameters.

The blue and red lines in (a) and (b) of Fig. 2 show the stationary operating states with constant air massflow ($\dot{m}_{\mathrm{air}} = 10.38$ g s^{-1}) and NG massflows ($\dot{m}_{\mathrm{NG}}$) between 0.46 and 0.53 g s^{-1} which will be considered in the remaining part of the paper. Plots (c), (d) and (e) of Fig. 2 compare the rms of the acoustic pressure, the time averaged temperature in the middle of the neck, and the resonance frequency of the damper for these constant air mass flow conditions. The resonance frequency of the damper is obtained from the transfer function relating the acoustic pressure in the combustion chamber to the one in the damper. The acoustic pressure in the combustion chamber for this transfer function is measured at the neck axial position, while the ones in the damper are recorded with a piezosensor located in its back volume. The cross power spectral density between the acoustic pressure in the damper and in the combustion chamber is divided by the power spectral density (PSD) of the acoustic pressure in the damper. This enables the isolation of the dynamics of the damper from the dynamics of the combustion chamber and extract its resonance frequency, as shown in Fig. 3. Fig. 2(e) reports the tuned and detuned damper frequency at the different operating conditions. The black dashed line represents the peak frequency of the combustor PSD at each condition when the damper is detuned (f_{peak}). As expected, the increase of the temperature in the neck of the damper is connected to a shift of damper resonance frequency away from the average f_{peak} when the damper is tuned.

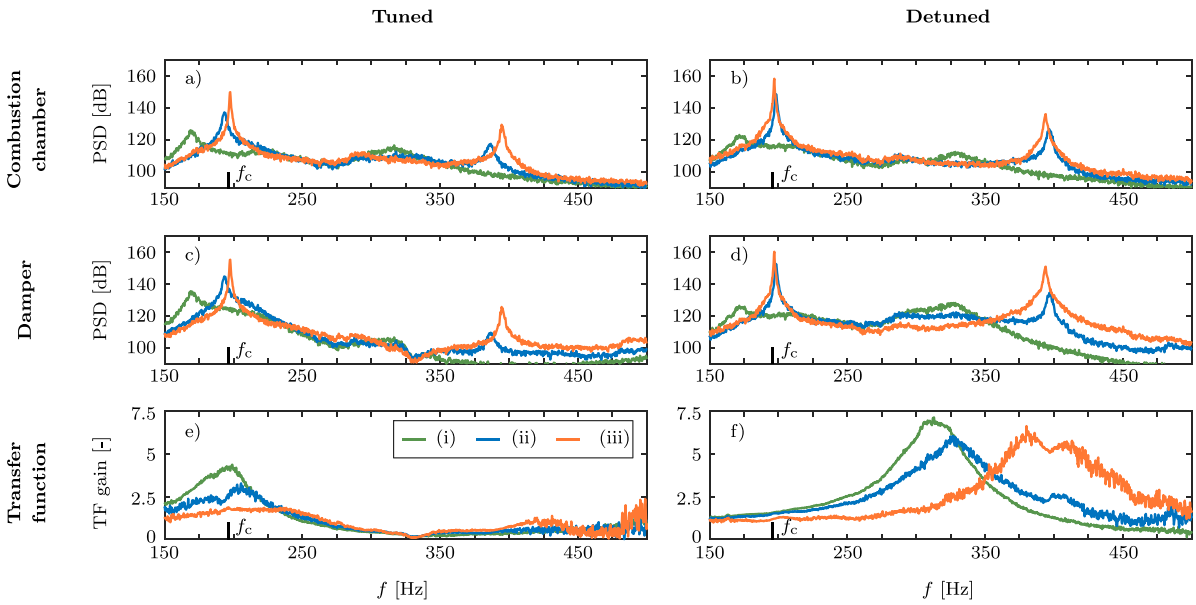

Fig. 3. Experimental PSD of the timetrace of a microphone placed in the combustion chamber at the damper axial position (a–b) and in the damper volume (c–d). (i)–(ii)–(iii) correspond to the three operating conditions highlighted in Fig. 2. (e) and (f) show the experimental transfer function, calculated as the ratio between the CPSD chamber–damper and the PSD in the damper (see Eq. (14)). (a)–(c)–(e) refer to the tuned damper, while (b)–(d)–(f) to the detuned case.

The distance between the two lines in plot (c) clearly shows the effect of the damper on the thermoacoustic stability border. The transition from linearly stable to unstable conditions in the tuned damper case is accompanied by a significant increase of the temperature in the neck as shown in Fig. 2(d) which is clearly not as strong for the detuned case. In fact, when the dominant mode of the combustion chamber forces the damper around its resonance frequency, the acoustic velocity amplitude in the neck becomes very large and is responsible of strong periodic ingestion of the hot gas. At these conditions, the resonance frequency of the damper changes accordingly, reducing its acoustic energy absorption at the frequency of the thermoacoustic instability. When the damper is detuned from the dominant mode of the combustor, the periodic ingestion of the hot gas is reduced because the damper is "forced" away from its resonance by the thermoacoustic mode, and therefore, the acoustic velocity in the neck is lower. Fig. 3 reports the PSDs of the signals recorded in the combustion chamber at the damper axial location (a–b) and in the damper volume (c–d). The operating conditions are (i)–(ii)–(iii) of Fig. 2 for the tuned (a–c) or detuned (b–d) damper. The acoustic energy of the PSDs in the combustor increase at peak frequency towards higher powers. Finally, (e) and (f) show the gain of the transfer function obtained as the ratio between the CPSD chamber–damper and the PSD in the damper volume. These plots isolate the damper response to the forcing of the combustion chamber, and capture the resonance frequency of the damper. The peak is shifted towards higher frequencies at (iii), when the ingestion of hot gas increases temperature in the neck.

The evolution of the probability density function (PDF) of the amplitude A of the Hilbert transform of the acoustic pressure signal P_∞ as function of the operating condition with constant air mass flow is shown in Fig. 4 (a). The corresponding time traces are shown in Fig. 4(b), (c) and (d) for $\dot{m}_{NG}$ = 0.46, 0.51 and 0.52 g s^{-1}, (i)–(ii)–(iii) of Figs. 2 and 3. Fig. 4(b) shows a linearly stable condition at low thermal power. In Fig. 4(c), the amplitude of the acoustic pressure exhibits erratic changes between two attractors, as highlighted by the two peaks of the PDF. Finally, Fig. 4(d) shows a linearly unstable case, where the presence of the damper is not able to prevent high amplitude thermoacoustic limit cycle in the combustor.

Let us now focus on the interesting condition of Fig. 4(c). The temperature timetrace measured in the middle of the neck is presented superposed to the acoustic pressure timetrace in the damper volume in Fig. 5. The acoustic signal is filtered with a bandpass filter around 196 Hz using a 40 Hz wide rectangular window function. For a better comparison with the 60 Hz temperature timetrace, the high-frequency oscillations of the amplitude of the pressure signal are smoothened with a moving average with a window of 1500 points. It is important to keep in mind that the thermal inertia of the thermocouple induces some delay in its response to changes of mean temperature induced by periodic hot gas ingestion. We find an optimum correlation between the delayed temperature and the acoustic amplitude for a delay τ = 1.37 s. From Figs. 4(c) and 5, we can infer the following scenario for explaining the erratic sudden changes of the acoustic amplitudes. During epochs of low acoustic amplitude, the periodic hot gas ingestion does not occur and the damper "operates" in the linear regime, with a resulting linear damping that keeps the coupled damper/combustor system linearly stable. Turbulent fluctuations of the heat release rate or hot gas flow in the vicinity of the neck can sporadically be sufficiently strong to perturb this marginal stability. Such a stochastic perturbation leads to the onset of self amplifying periodic hot gas ingestion events that bring the system to a limit cycle state. The self amplification is due to the alternation of the resistance and of the reactance of the damper when it is subject to hot gas ingestion of increasingly amplitude. This scenario is modelled and validated in the next section where we show the key importance of (i) the damper response nonlinearity when it

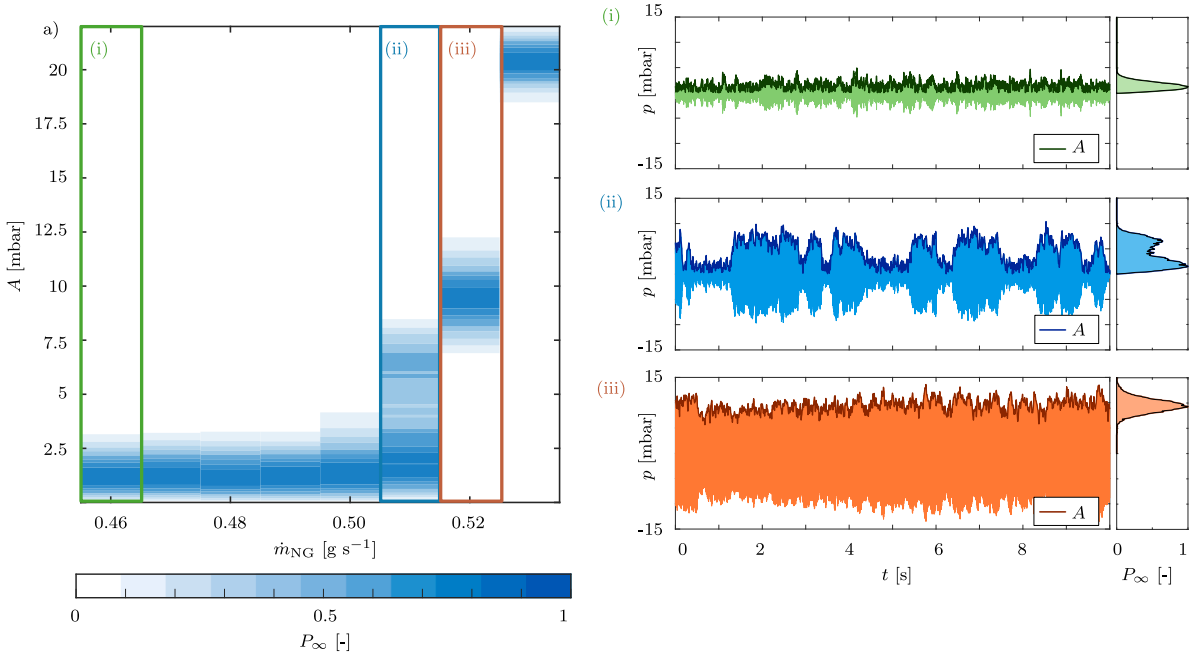

Fig. 4. Experimentally measured transition from stable to unstable thermoacoustic conditions (red line of Fig. 2). The map shows the PDF (P_∞) of the pressure oscillations amplitude A in the combustor at the damper's location. On the right, the time traces of the pressure oscillations p and the PDF of their amplitude A of the three points highlighted on the map are used to determine their thermoacoustic condition: stable (i), bistable (ii) and unstable (iii).

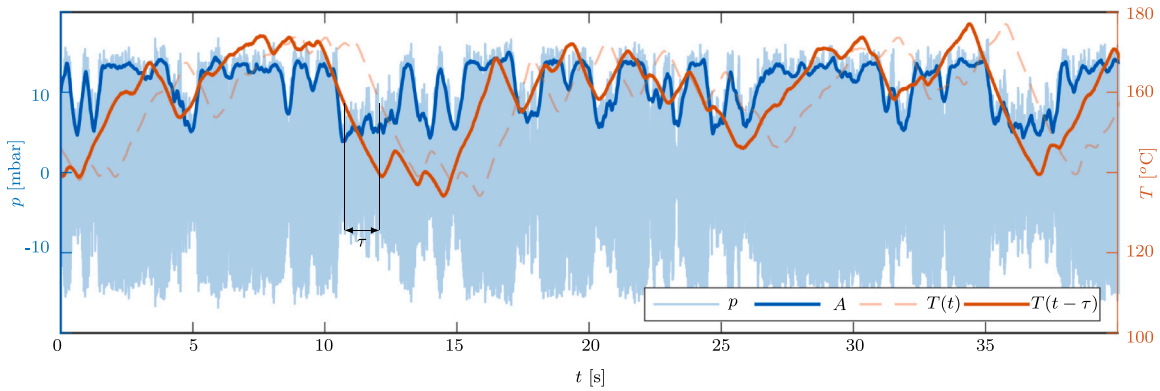

Fig. 5. Timetraces of the acoustic pressure in the damper volume superposed to the temperature in the neck at earlier time τ = 1.37 s. The thick blue line represents the envelope of the acoustic pressure signal smoothened with a 1500 points window moving average. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

is subject to acoustic amplitudes that are sufficiently large to trigger periodic hot gas ingestion and (ii) the stochastic forcing from the turbulent reactive flow which enables the intermittent escapes from and returns to the low acoustic amplitude attractors.

3. Model

This section introduces a low order model describing the coupled system damper–combustor. In the first subsection, we will show that the damper can be described as a Duffing-type oscillator when the phenomenon of the periodic hot gas ingestion is considered. Subsequently, it will be coupled to a Van der Pol oscillator which describes the dominant thermoacoustic mode of the combustion chamber. The parameters of the ordinary differential equation describing the damper dynamics are calibrated from the experimental data. The results presented in the following sections are obtained by running a 60 s Simulink simulation with a fixed-step ode5 (Dormand–Prince) solver of the calibrated model. The timestep of the solver is fixed to be 0.001 f_c^{-1} where f_c is the resonance frequency of the combustor (fixed at 196 Hz in the model simulations).

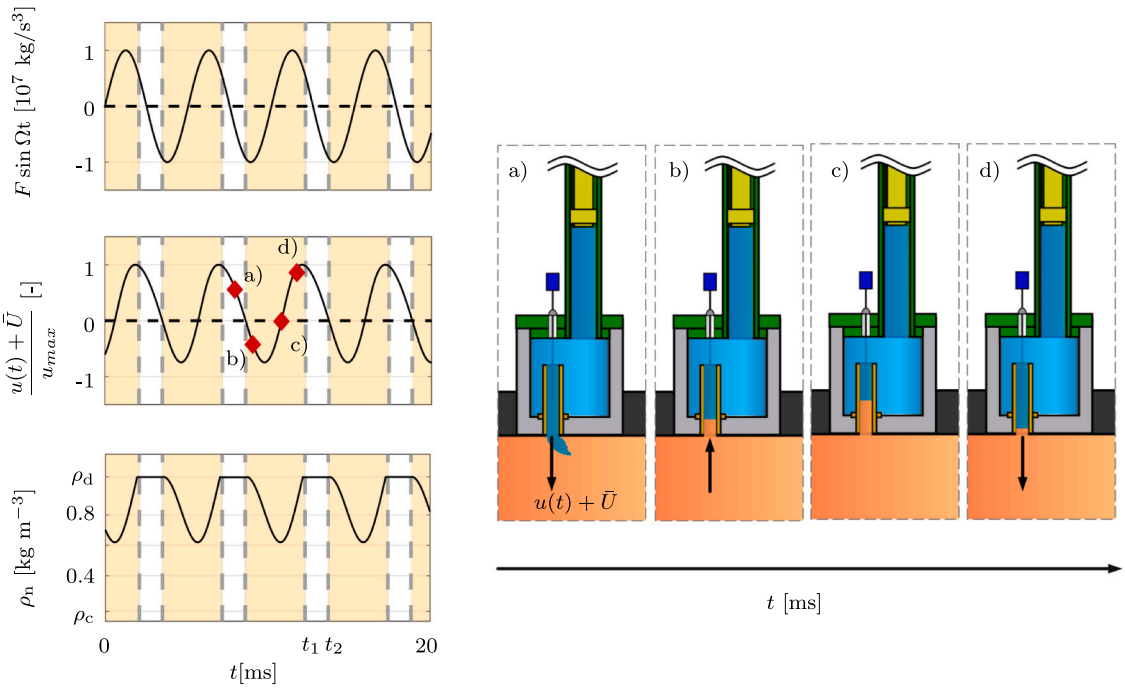

Fig. 6. Forcing F, velocity oscillations in the neck normalized by their maximum $(u(t)+\bar{U})/u_{\max}$ of the damper and mean density in the neck of the damper ρ_n. On the right, a sketch of the damper highlights the presence of the hot gas in the neck and the direction of the velocity for three conditions on an acoustic cycle (a), (b), (c) and (d). Values used in Eqs. (2) to (5) for generating the timetraces are: ζ = 1.5, $\bar{U}$ = 4.2 m s^{-1}, Ω = 1.231 rad s^{-1}, F = 10^7 m s^{-3}, l_{eff} = 5.2 × 10^{-2} m, ρ_{d} = 1.05 kg m^{-3}, ρ_{c} = 0.16 kg m^{-3}, c_{d} = 366 m s^{-1}, V_{d} = 1.8 × 10^{-4} m^3, a = 1.13 × 10^{-2} m^2.

3.1. Model of the damper

As indicated by Eq. (1), the speed of sound of the gas inside a Helmholtz resonator and its geometry define its resonance frequency. A simplified description of this acoustic resonator when it is purged with a mean flow is a nonlinearly damped harmonic oscillator:

$$\ddot{u} + \frac{\zeta}{l_{\text{eff}}}|\overline{U} + u|\dot{u} + \Omega_{\text{d}}^2(t)u = F\sin(\Omega t), \tag{2}$$

where u is the acoustic velocity in the neck as derived in [10,28,35]. Note that by convention the velocity is considered positive if directed towards the combustion chamber. In Eq. (2), ζ represents the pressure loss coefficient, l_{eff} is the effective length of the neck [11], $\overline{U}$ is the mean purge flow velocity in the neck and F the amplitude of the forcing, proportional to the time derivative of the acoustic pressure in front of the damper neck. When the acoustic amplitude exceeds the mean flow velocity $\overline{U}$, periodic reversals of the flow occur and the stiffness becomes nonlinear with respect to the acoustic forcing amplitude. The periodic reversals of the flow are accompanied with periodic hot gas ingestion (PHGI).

The mean variation of the density in the neck due to PHGI affects the tuning of the damper according to:

$$\Omega_{\text{d}}(t) = c_{\text{d}}\sqrt{\frac{a}{V_{\text{d}}l_{\text{eff}}}}\sqrt{\frac{\rho_d}{\rho_n(t)}}, \tag{3}$$

where c_{d} and ρ_{d} are the speed of sound and the air density in the back volume of the damper and where ρ_{n} is the mean density in the neck, which varies in time in case of PHGI. Eq. (3) is derived from the linearized mass and momentum balance in the neck of a Helmholtz damper, with the assumption that the acoustic pressure oscillations are uniform in the damper volume, and by accounting for the temporal variation of averaged density in the neck induced by the temperature oscillations.

When there is no PHGI, i.e. $u(t) < \overline{U}$, $\rho_{\text{n}}(t) = \rho_{\text{d}}$ and this formula simplifies to $\Omega_{\text{d}} = \omega_{\text{d}}$ given in Eq. (1). For PHGI to happen, the acoustic velocity amplitude must exceed the mean flow velocity. In such situation, for each acoustic cycle, there is an interval $[t_1, t_2]$ during which the neck is partially filled with hot gas (see Fig. 6). The relation between the acoustic velocity u and the instantaneous mean density in the neck of the damper $\rho_n(t)$ has been first proposed in [28]:

$$\rho_n(t) = \begin{cases} \rho_{\text{d}} - \left(\rho_{\text{d}} - \rho_{\text{c}}\right)\alpha(t) & \text{if } t_1 < t < t_2 \\ \rho_{\text{d}} & \text{if } t_2 < t < t_1 + \frac{2\pi}{\omega} \end{cases} \tag{4}$$

where

$$\alpha(t) = \frac{1}{l}\left|\int_{t_1}^{t}(\overline{U}+u)dt\right| \tag{5}$$

and where the density of the gas in the combustion chamber is denoted by ρ_c. As previously mentioned, the PHGI is limited by two events. Firstly, a reversal of the total velocity in the neck, i.e. $(u+\overline{U})$ becomes negative and the hot gas starts entering the neck. Secondly, the complete removal of hot gas, i.e., the moment when the hot gas is completely flushed out of the neck. The time of occurrence of t_2 is calculated as the first zero in time of the acoustic displacement after t_1, i.e., $\int_{t_1}^{t_2}(\overline{U}+u)dt = 0$. Further details on their derivation can be found in [28].

The function $\alpha(t)$ represents the ratio between the acoustic displacement of the idealized interface between the purge air and ingested hot gas and the physical length of the neck of the Helmholtz resonator l. The model is limited to situations where the acoustic velocity amplitude in the neck is not sufficiently large to induce hot gas penetration in the back volume, i.e. the maximum displacement of the "interface" separating purge air and hot gas is smaller than the neck length.

The three plots of Fig. 6 show the amplitude of the forcing term, the velocity and mean density oscillations in the neck of the damper, obtained from Eqs. (2) to (5). The forcing term is a sine wave with amplitude $F = 10^7$ m s^{-3} and frequency equal to the resonance frequency of the damper $\Omega = \omega_d$. The numerical value of the geometrical parameters, the damper's resonance frequency $\omega_d = 1231$ rad s^{-1}, the mean velocity in the neck $\overline{U} = 4.2$ m s^{-1}, which were used for producing the time traces of Fig. 4 are indicated in the caption of that figure.

The velocity oscillations in the neck of the damper are asymmetric with respect to the mean velocity during a single cycle. When the total velocity $(\overline{U}+u)$ in the neck of the damper is negative, the mean density in the neck decreases (sketch c) of Fig. 6. When the velocity in the neck changes its sign, it starts to expel the hot gas again.

3.2. Duffing–Van der Pol oscillator

Eq. (2) models the velocity oscillations in the neck of the damper as a harmonic oscillator with nonlinear damping and stiffness. The forced oscillations are analysed by selectively suppressing the different nonlinearities. In the analysis, the following 3 equations will be considered:

$$\ddot{u} + \frac{\zeta}{l_{\text{eff}}}|\overline{U}+u|\dot{u} + \omega_d^2 u = F\sin(\Omega t) \qquad \text{(nonlinear damping, linear stiffness)} \tag{6}$$

$$\ddot{u} + \frac{\zeta}{l_{\text{eff}}}|\overline{U}|\dot{u} + \Omega_d^2(t)u = F\sin(\Omega t) \qquad \text{(linear damping, nonlinear stiffness)} \tag{7}$$

$$\ddot{u} + \frac{\zeta}{l_{\text{eff}}}|\overline{U}|\dot{u} + \omega_d^2 u = F\sin(\Omega t) \qquad \text{(linear terms only)} \tag{8}$$

Fig. 7 shows the amplitude B of the acoustic velocity oscillations u in the damper neck at the forcing frequency Ω for different forcing amplitudes F and for different variants of the model given by Eqs. (2), (6), (7) and (8).

In order to detect the hysteresis, two computations were carried out, first by ramping up the forcing frequency Ω, and second by ramping it down.

As expected, the presence of nonlinear damping in Eq. (6) reduces the amplitude of the response around the resonance frequency at high forcing amplitude, which can be seen in Fig. 7(b). When only the nonlinear stiffness term caused by the hot gas ingestion is considered, i.e. Eq. (7), the behaviour of the damper's response is similar to the one of a Duffing oscillator. In fact, the maximum response is shifted towards a higher frequency and a hysteresis is generated, as shown in Fig. 7(c). When both the nonlinear damping and stiffness are considered, i.e. Eq. (2), the response of the damper takes on some features of the previous cases (see Fig. 7(d)). In fact, the nonlinearity in the damping term still reduces the amplitude of the response, but this time the maximum response amplitude moves towards higher frequencies due to the effect of the nonlinear stiffness term. For the given set of parameters, the effect of nonlinear damping is prevailing on the onset of the Duffing-like hysteresis.

3.3. Model of the coupled system

The system damper–combustor is modelled as a system of coupled nonlinear oscillators:

$$\ddot{\eta} - (2\nu + \kappa\eta^2)\dot{\eta} + \omega_c^2\eta = \frac{\varepsilon a \rho_d c_d^2}{\Psi V_d}\dot{u} + \chi(t) \tag{9}$$

$$\ddot{u} + \frac{\zeta}{l_{\text{eff}}}|\overline{U}+u|\dot{u} + \Omega_d^2(t)u = -\frac{\Psi}{\rho_n(t)l_{\text{eff}}}\dot{\eta}. \tag{10}$$

The first equation is derived from the model considered and experimentally validated in [36]. It describes the dynamics of the dominant thermoacoustic mode oscillating at ω_c in the combustion chamber, when it is coupled, via $\dot{u}$, to a nonlinear damper described by the second equation. These coupled equation have been proposed in [35] for an isothermal Helmholtz damper with constant resonance frequency, i.e. for $\Omega_d(t) = \omega_d$. In that reference, the model has been validated with an isothermal electroacoustic experimental setup. The key novelty of the present work is to include the dynamic detuning of the Helmholtz damper due to the

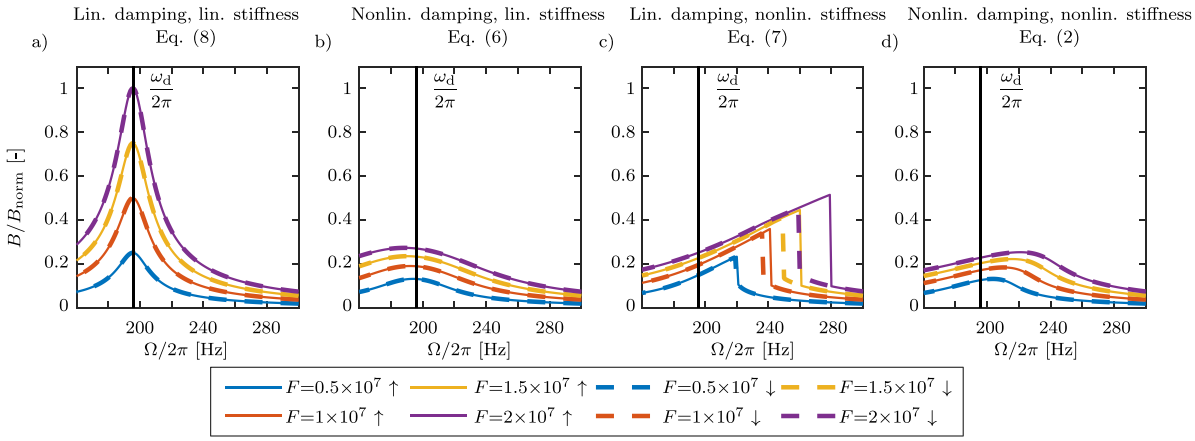

Fig. 7. Amplitude of the velocity oscillations in the neck of the damper B normalized by its maximum among the different cases (B_{norm}) obtained when Eqs. (2), (6), (7) and (8) are used.

varying density in its neck induced by PHGI, i.e. for $\Omega_{\mathrm{d}}(t)$ given by Eqs. (3)–(4)–(5). Let us further consider Eq. (9). The amplitude of the dominant thermoacoustic mode η is modelled as a nonlinear stochastic oscillator:

$$\ddot{\eta} + \omega_{\mathrm{c}}^2 \eta = f(\eta, \dot{\eta}) + \chi(t). \tag{11}$$

In Eq. (11), χ is the additive turbulence-induced stochastic forcing and, as in [36], the function $f(\eta, \dot{\eta}) = \dot{q} - \alpha\dot{\eta}$ results from the balance between the nonlinear flame response $\dot{q}$ and the linear acoustic damping α. The flame response is expanded up to the third order, including a cubic saturation term as $q = \beta\eta - \kappa\eta^3/3$. The linear growth rate ν is defined as $(\beta - \alpha)/2$ and defines the linear stability of the combustion chamber. When $\beta > \alpha$ the acoustic perturbations will be amplified at the growth rate $\nu > 0$. As a result, the self-sustained thermoacoustic oscillations in the combustor are modelled as a Van der Pol oscillator for the acoustic pressure driven by additive noise. In Eqs. (9)–(10), the mode amplitude at the damper axial position is defined as Ψ, and ε represents a damping efficiency factor, defined in Eq. (12):

$$\varepsilon = \frac{V_{\mathrm{d}}\Psi^2}{V_{\mathrm{c}}\Lambda}. \tag{12}$$

Here, V_{c} is the volume of the combustor. The coefficient Λ represents the norm of the mode:

$$\Lambda = \frac{1}{V_{\mathrm{c}}} \int_{V_{\mathrm{c}}} |\psi|^2 dV, \tag{13}$$

where ψ is the shape of the longitudinal eigenmode involved in this work. In the present Λ will be assumed to be 1/2, which is the theoretical value for 1 dimensional eigenmodes in a constant cross-sectional duct. This approach assumes the spatial distribution of eigenmode to be one dimensional in the combustion chamber, i.e. it corresponds to a longitudinal mode, and neglects the effect of density change across the compact flame [9]. In addition, the influence of the damper on the modeshape is neglected due to the large volume difference between large (the combustion chamber) and the small (the damper) cavity [25]. In order to ease the time marching of the system of coupled ODEs, the additive forcing χ which can be considered as a white noise is modelled as an Ornstein–Uhlenbeck (OU) process [42] with correlation time that is very short compared to the acoustic period. It is governed by $\dot{\chi} = -\frac{\chi}{\tau_\chi} + \zeta_\chi$, where τ_χ is the correlation time of the OU noise and ζ_χ is Gaussian white noise with intensity Γ_χ/τ_χ^2. The correlation time is fixed to 0.1 ms. Thus, in the frequency range of interest (i.e. 100–500 Hz) the PSD of the OU noise is constant, as for a white noise. The equation for the acoustic velocity in the damper neck has a forcing term with the time derivative of the acoustic pressure in the combustor in front of the neck of the damper.

3.4. Identification of the model parameters

In the following section, the parameters of Eqs. (9)–(10) are calibrated with the experimental measurements described in Section 2. The identified parameters will be employed to investigate the ability of the model to reproduce the nonlinear dynamics observed in the experiments.

Several parameters are readily given by the geometry or the operating condition of the experimental configuration. They are summarized in Table 1:

Table 1
Geometrical parameters of the system damper-combustion chamber.

V_c [m^3]	V_d [m^3]	a [m^2]	ρ_d [kg m^{-3}]	c_d [m s^{-1}]
5.2×10^{-3}	1.8×10^{-4}	1.1×10^{-4}	1.05	354

3.4.1. Parameters of the damper equation

The pressure loss coefficient ζ and the effective length l_{eff} can be estimated by fitting the transfer function between acoustic pressure in the combustion chamber (input) and the one in the volume of the damper (output). To safely use this fitting approach, only data collected for linearly stable conditions must be considered. This is because the transfer function of interest is linear. In fact, when the system is thermoacoustically stable, the amplitude of the acoustic pressure in the combustion chamber is low, the nonlinear terms in Eq. (10) can be neglected.

The acoustic signals are measured by a microphone close to the neck of the damper in the combustion chamber and by a piezosensor in the back volume of the damper. The experimental transfer function was then calculated as:

$$H = \frac{\hat{S}_{\mathrm{cd}}}{\hat{S}_{\mathrm{cc}}}, \tag{14}$$

where $\hat{S}_{\mathrm{cd}}$ is the cross-power spectral density (CPSD) between the signal in the combustor and in the damper. $\hat{S}_{\mathrm{cc}}$ is the power spectral density (PSD) of the acoustic pressure in the combustion chamber. Note that the dominant source of dynamic noise in the system is the combustion noise produced by the turbulent fluctuations of the heat release rate in the combustor. Consequently, since it is not in the damper, applying the same approach for identifying the parameters of the left hand side of Eq. (9) would not work, even though the coupled system is linearly stable.

The model used for the fit of the transfer function is obtained by linearizing Eqs. (9) and (10) and by using the following equation:

$$u = -\frac{V_{\mathrm{d}}}{c_{\mathrm{d}}^2 \rho_{\mathrm{d}} a} \dot{p}_{\mathrm{d}}, \tag{15}$$

where u represents the velocity oscillations in the neck of the damper, positive when pointing out from the damper and P_{d} is the acoustic pressure in the volume of the damper.

The linear transfer function linking velocity fluctuations in the neck to acoustic pressure in the combustion chamber is:

$$\frac{\hat{u}}{\Psi \hat{\eta}} = -\frac{\dfrac{s}{\rho_{\mathrm{d}} l_{\mathrm{eff}}}}{s^2 + \dfrac{\zeta}{l_{\mathrm{eff}}} \overline{U} s + \omega_{\mathrm{d}}^2}, \tag{16}$$

where $s = \mathrm{i}\omega$ is the Laplace variable. Combining the Laplace transform of (15) with (16), we can obtain the linear transfer function linking the pressure oscillations in the volume of the damper (P_{d}) and the ones in the combustion chamber at the location of the damper:

$$\frac{\hat{p}_{\mathrm{d}}}{\Psi \hat{\eta}} = \frac{\dfrac{c_{\mathrm{d}}^2 a}{V_{\mathrm{d}} l_{\mathrm{eff}}}}{s^2 + \dfrac{\zeta}{l_{\mathrm{eff}}} \overline{U} s + \omega_d^2}. \tag{17}$$

The parameters are then identified by fitting this transfer function model to the experimental one for thermoacoustically stable conditions with the Fast Relax Vector Fitting routine, developed by Gustavsen in [43]. The results from this fit are shown in Fig. 8 in terms of the transfer function gain. As indicated above, this parameter estimation approach cannot be used when the system dynamic is nonlinear, and therefore it was only applied for $\dot{m}_{\mathrm{NG}}$ ranging between 0.46 and 0.5 g s^{-1}. Once the estimates of the pressure loss coefficient ζ are obtained, the effective length of the neck of the damper can be deduced from Eq. (1) knowing the ambient temperature and the geometrical shape of the damper. Plot (b) and (c) of Fig. 8 highlight the evolution of l_{eff} and ζ over the considered operating conditions. Their small variation allows the use of their mean in the remainder of this study, 0.05 m and 2.9, respectively. Moreover, we assume that with these values, the model of Eq. (2) correctly predicts the nonlinear damper response when the latter is forced at high amplitude. This assumption has been validated experimentally in [35] for the simpler case of periodic flow reversal, which is the isothermal counterpart of the PHGI.

3.4.2. Parameters of the combustion equation

The last parameters to be identified are the ones related to the thermoacoustic dynamics of the combustor. These parameters are: the growth rate ν and the saturation coefficient κ of the Van der Pol oscillator used to represent the saturation of the flame response to high amplitude acoustic feedback, the intensity of the additive noise Γ_χ and the amplitude of the mode at the axial position of the damper Ψ. As it was shown in Section 2.2, the presence of the damper has a strong impact on the nonlinear dynamics of the system. This effect precludes the use of the method presented in [44] to identify the linear growth rate of the system. This is because obviously, in the present configuration the dynamics cannot be reduced to a single self-sustained oscillator and must also include

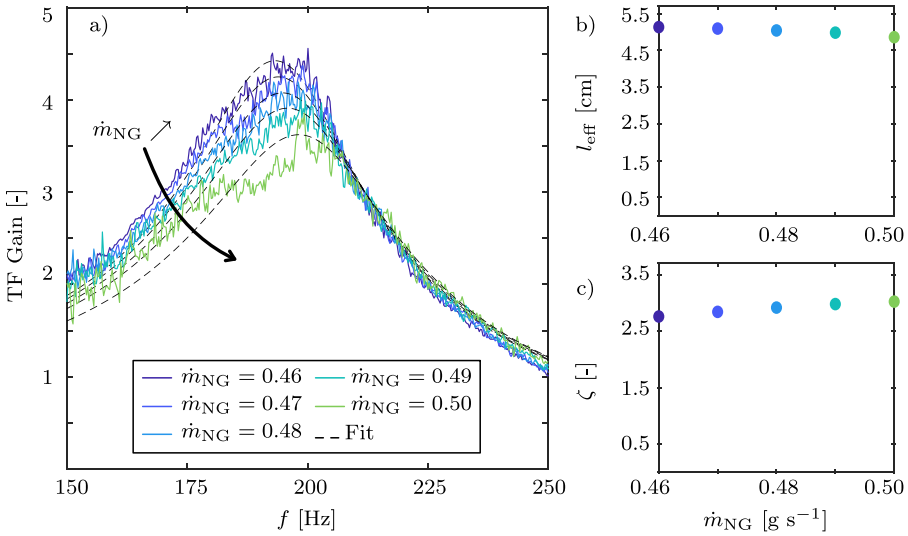

Fig. 8. (a) Gain of the experimental transfer function between the microphone in the combustion chamber at the axial location of the damper and the one in the damper volume. The dashed back lines show the results of the fittings. The effective length l_{eff} and the pressure loss coefficient ζ obtained with the fitting routine for the different operating conditions are shown in (b) and (c).

the feedback from the nonlinear damper. More details about the applicability of the method presented in [44] for a single nonlinear oscillator are given in [45,46]. Nevertheless, as indicated above, most of the parameters of the coupled oscillator model given in Eqs. (9) and (10) have been reliably identified, either because they are defined by the geometry and the operating conditions, or because they could be robustly quantified from acoustic time series where the system was operated at thermoacoustically stable conditions. The few remaining ones are now calibrated by considering the available observables. First, with the set of microphones, the axial distribution of the acoustic mode $\Psi(x)$ and the mode amplitude $\eta(t)$ could be estimated, allowing us to identify $\Psi = 0.7$. Knowing the amplitude of the limit cycles with detuned damper being of the order of 10 mbar for unstable conditions, the ratio ν/κ could be guessed. Indeed, with a detuned damper the limit cycle amplitude of the model is $\sqrt{\nu/\kappa}$. The positive growth rates identified in a similar configuration, but for a different operating conditions and without damper [39] were about 10 rad s^{-1}. Testing the model with such growth rates did not allow us to reproduce the bistability of Fig. 4c. However, we found that ranging from 45 to 70 rad s^{-1} allows us to fully retrieve the observed nonlinear dynamics. By fixing this range, a value of $\kappa = 0.001$ 1/Pa^{-1} is readily obtained and allows us to obtain the right amplitude in Pa. Finally, it is found that $\Gamma_\chi = 4.6 \times 10^{12}$ Pa s^{-1} is optimum for retrieving both the PDF of the acoustic envelope and the frequency of sporadic amplitude bursts in the vicinity of the bifurcation.

4. Results

In this section, the results obtained with the model will be presented and commented. In the first subsection, the dimensions of the neck of the damper are the same as in the experiments, in order to compare the model against the experimental results. In the second subsection, the diameter of the damper neck is virtually increased, in order to further investigate the nonlinear dynamics of the coupled system.

As a summary, the main assumptions employed in the derivation the model are now listed:

- The acoustic oscillations are represented by a single dominant mode
- The nonlinearities of the flame response are approximated with a cubic saturation term
- Length and diameter of the damper neck and damper volume are compact with respect to the mode wavelength
- The density variation induced by the hot gas ingestion is modelled as a time-dependent and spatially-averaged equation

4.1. Comparison of the calibrated model and the experiments

The system with detuned damper features the typical characteristics of a supercritical Hopf bifurcation when the fuel mass flow ($\dot{m}_{NG}$) is increased. Moreover, as explained in Section 3.4.2, when the linear growth rate ν in absence of damper is varied between 40 and 70 rad/s for a fixed nonlinearity constant, the observed stochastic dynamics is very well reproduced by the model. Therefore, as a first approximation, this monotonic increase of the linear growth rate as function of the fuel mass flow is considered linear and the other parameters of the model are kept constant. It is important to note that such linear evolution is not valid in general. Indeed, it is well known that thermoacoustic stability properties of a combustion system evolve non-monotonically with system parameters when wide ranges of operating conditions are explored, e.g. [39].

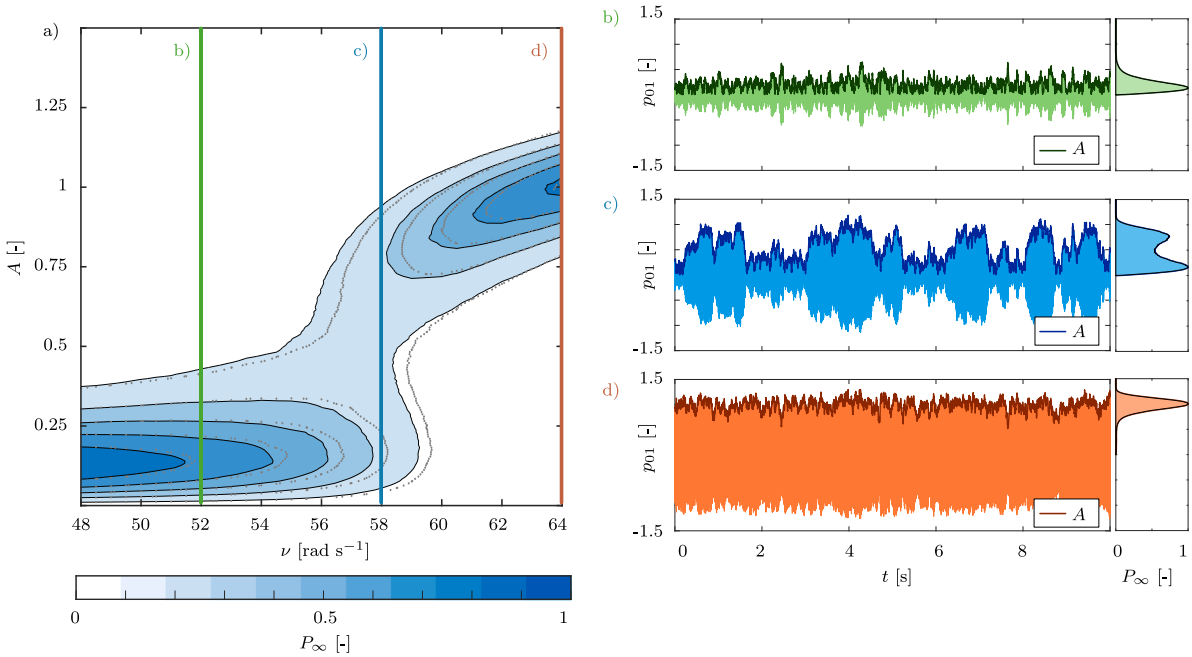

Fig. 9. Map of the stationary PDF (P_∞) of the amplitude of the acoustic pressure A as function of the linear growth rate ν. The acoustic pressure oscillations are normalized by the amplitude of the most probable acoustic amplitude at the largest linear growth rate ν. P_∞ is obtained from time domain simulations of Eqs. (9) and (10), which are sufficiently long for convergence to a smooth PDF. (b), (c) and (d) show the time traces normalized by the same value and the PDF of A for three significant cases, respectively: linearly stable, bistable and linearly unstable thermoacoustic conditions. It is important to underline the similarity between the three PDFs of the simulated time traces and the ones recorded in the experiments (see b–c–d of Fig. 4). The dashed contour in (a) shows P_∞ when the stiffness nonlinearity is suppressed by assuming $\Omega_d = \omega_d$ in Eq. (10). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The coloured contourplot of Fig. 9 shows the stationary PDF of the amplitude A of the Hilbert transform of the acoustic pressure in the combustion chamber (P_∞). This PDF is obtained by performing time domain simulations of the model given by Eqs. (9) and (10) for fixed values of the bifurcation parameter ν and for simulated time that are sufficiently long to obtain smooth and converged P_∞. Short time intervals from these long simulated periods are presented in Fig. 9 for typical linear growth rates as previously done for the experimental results. Comparing Figs. 4 and 9, it is clear that the model reproduces very well the features of the PDF and of the time series of the acoustic pressure measured in the combustor.

In order to get further insights about the relative contributions of the nonlinear damping and nonlinear stiffness of the coupled system, other simulations were performed with the constraint of fixing the damper's resonance frequency, i.e. $\Omega_d(t) = \omega_d$. Indeed, the dashed lines in Fig. 9 shows the contours of the stationary PDF for a coupled system which does not account for the variation of stiffness caused by the hot gas ingestion. This system has otherwise the same parameters identified in the previous sections. The density in the neck is considered to be always constant and equal to the one in the back volume of the damper. The comparison with the filled contourplot shows that the difference introduced by the oscillations of the mean density in the neck caused by the periodic hot gas ingestion does not significantly impact the nonlinear dynamics in the considered damper geometry.

In fact, the strong damping nonlinearity drowns out the one introduced by the stiffness modulation induced by the PHGI. The damper designed for these experiments exhibits a resonance peak which is significantly wider than the peak of the governing thermoacoustic mode, which explains the predominance of the nonlinear damping induced by the PHGI upon the nonlinear dynamics of the coupled system (see Fig. 8). In the next section, a damper with a smaller damping term will be studied with the model and compared to the one employed in the present experiments.

4.2. Damper neck geometry variation

The dimensions of the neck of the damper used in the experiments are now modified in the model in order to further study the influence of the density modulation in the neck of the damper during periodic hot gas ingestion. In particular, the diameter of the cross section of the neck is increased by 20% (from 12 to 14.4 mm). The effective length of the damper is varied accordingly (from 52 to 75 mm) in order to keep the resonance frequency constant for the same back volume of the damper. The parameters of the two damper necks are summarized in Table 2. They will be referred as Damper 1 (D1) and Damper 2 (D2). D1 corresponds to the damper of the experiments. The purge massflow in D2 is assumed to be the same as in D1, therefore the increase of the neck cross-sectional area is associated with a reduction of the purge flow velocity. All the other parameters of the coupled system given in Eqs. (9) and (10) are identical to the ones of the previous section.

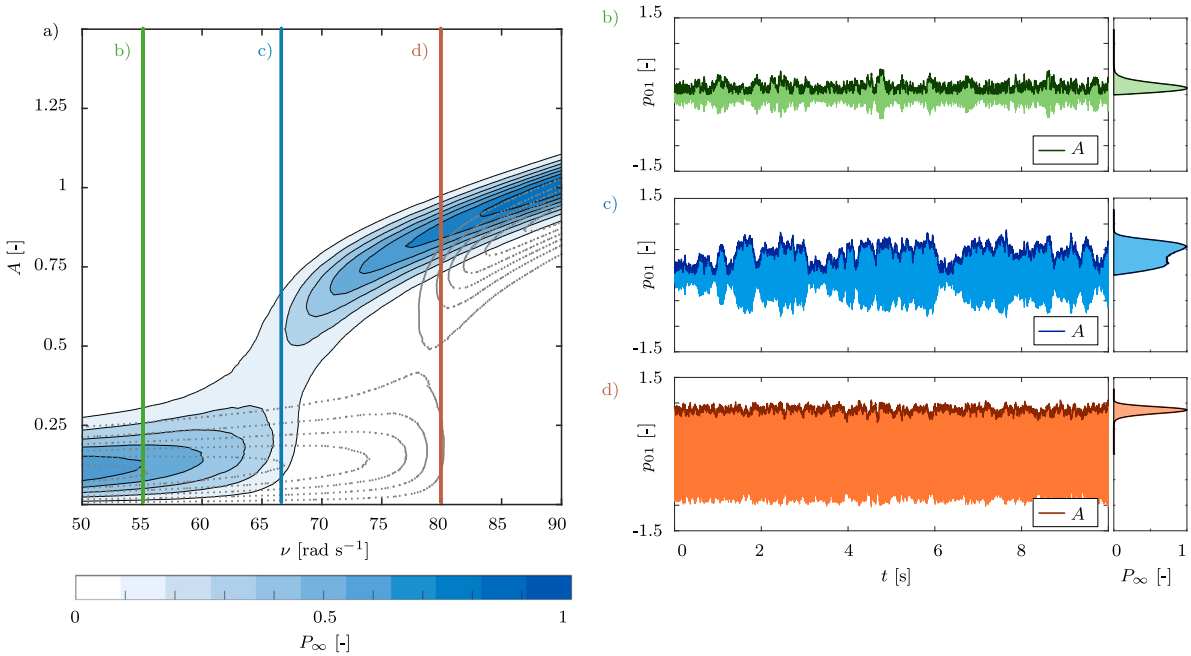

Fig. 10. (a) Stationary PDF (P_∞) of the amplitude of the acoustic pressure A as function of the linear growth rate ν for damper D2. The acoustic pressure oscillations are normalized by the amplitude of the most probable acoustic amplitude at the largest linear growth rate conditions (A). P_∞ is obtained from time domain simulations of Eqs. (9) and (10), which are sufficiently long for convergence to a smooth PDF. (b), (c) and (d) show the time traces and the PDF of three significant cases: linearly stable, bistable and linearly unstable thermoacoustic conditions. The dashed contour in (a) shows P_∞ obtained suppressing the nonlinearity by assuming $\Omega_d = \omega_d$ in Eq. (10).

Table 2
Comparison of the parameters between the two damper necks.

Parameters	Damper 1	Damper 2
$\sqrt{4a/\pi} = d$ [mm]	12	14.4
l_{eff} [mm]	52	75
$\overline{U}$ [m s^{-1}]	4.2	2.9

As it was done in the previous section, Fig. 10 compares the contours of the PDF of the acoustic amplitude as function of the linear growth rates ν for D2, accounting or not in the model for the nonlinear stiffness due to PHGI.

The effect of the nonlinear stiffness now heavily affects the shape of P_∞ and the position of the bistable region in the bifurcation diagram. It is observed that a nonlinear stiffness in the damper equation of the coupled system (10) shifts the bistable zone, which is the transition from linearly stable thermoacoustic conditions to thermoacoustic limit cycles, to smaller growth rates. The three plots (b), (c) and (d) show three time traces of the pressure oscillations and the normalized stationary PDFs of their amplitudes, when accounting for the nonlinear stiffness.

The forced response of dampers D1 and D2 are now computed. This analysis is motivated by the fact that D2 features a nonlinear stiffness that influences the coupled system dynamics to a larger extent than the one of D1. Fig. 11 shows the mean amplitude of the acoustic velocity in the neck of the damper for different forcing frequencies $\Omega/2\pi$ when Eqs. (2) and (6) are used for the parameters of D1 and D2. As the mean purge mass flow is kept constant between the two cases, the increase of the cross-sectional area of the neck D2 is accompanied with a mean purge flow velocity $\overline{U}$ reduction in its neck, resulting in a damping decrease. Therefore, for a given forcing amplitude F, the amplitude of the velocity oscillations u of the forced damper is increased regardless of the phenomenon of periodic hot gas ingestion when switching from D1 to D2. At the same time, the reduction of the damping term makes the effect of the stiffness nonlinearity induced by the hot gas ingestion more pronounced. In fact, a larger acoustic velocity u reduction connected with the introduction of the dynamic detuning (from Eqs. (6) to (2)) is observed for D2 compared to D1 at the resonance frequency of the combustion chamber (f_c in Fig. 11) at constant forcing amplitude F.

As a consequence, the transition of the coupled system from linearly stable to limit cycle thermoacoustic conditions occurs at lower growth rates. The nonlinearity of the damping does not drown out the nonlinear stiffness induced by PHGI anymore and its effect could be isolated. This result shows the importance of carefully accounting for nonlinear damping and nonlinear stiffness in the design phase of a damper.

The two same colour lines show the solutions of Eq. (2) (full line) and Eq. (6) (dashed line) with a starting point u= 0.1 m s^{-1}, calculated for different forcing frequencies $\Omega/2\pi$. The black vertical line highlights $\omega_c/2\pi$, the resonance frequency of the damper.

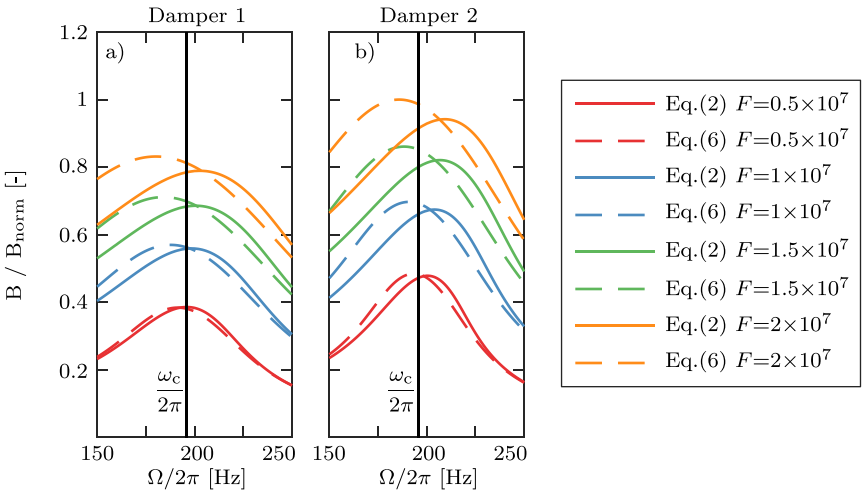

Fig. 11. Amplitude of the acoustic velocity in the neck of the damper B when accounting or not for the nonlinear stiffness caused by the PHGI for dampers D1 (a) and D2 (b). The value on the y axis is normalized by B_{norm}, the maximum acoustic velocity amplitude among all the presented cases in (a) and (b).

5. Conclusions

In this paper, the phenomenon of the hot gas ingestion, which can occur in the neck of a Helmholtz damper used for suppressing thermoacoustic instabilities, has been investigated experimentally and reproduced with a low order model. It was shown that the bistable region separating linearly stable conditions and thermoacoustic limit cycles in the bifurcation diagram is governed by (i) the nonlinear damping and the nonlinear stiffness induced by PHGI and (ii) the intensity of the stochastic forcing of the thermoacoustic system from turbulence. The key finding of this work is that the dynamic detuning due to the modulation of the mean density of the resonator neck is not necessarily more influential on the nonlinear dynamics than the associated damping nonlinearity. Indeed, the respective contribution of nonlinear damping and nonlinear stiffness on the failure of a damper to prevent thermoacoustic instabilities strongly depend on the quality factor of the damper.

This work not only demonstrates the crucial need for considering these nonlinearities during the design of such dampers, but also provides a validated low order model that fully reproduces the stochastic nonlinear dynamics of a coupled system damper–combustor.

CRediT authorship contribution statement

Luigi Miniero: Investigation, Methodology, Validation, Software, Formal analysis, Visualization, Writing – original draft. **Georg A. Mensah:** Software, Validation. **Claire Bourquard:** Investigation, Methodology. **Nicolas Noiray:** Conceptualization, Investigation, Methodology, Supervision, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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